Ruben Van Paemel, MD/PhD

Pediatrician with experience in pediatric oncology, hematology and precision medicine, with a PhD in cancer genomics, liquid biopsy biomarkers, big data analysis and artificial intelligence.

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github.com/rmvpaeme

EXPERIENCE & EDUCATION

Oct 2024 – Present

Fellow in Pediatric Hematology & Oncology

Ghent University Hospital, Ghent, Belgium

- Diagnosis and treatment in hematology, oncology and stem cell transplantation
- Management of chemotherapy complications in a multidisciplinary team
- Enrolment of patients in phase I/II clinical trials (ICF, safety monitoring, adverse event reporting)

June 2017 – 2024

Residency & Master in Specialist Medicine – Pediatrics

Ghent University & AZ Sint-Lucas, Ghent, Belgium

7-year specialist training programme covering general pediatrics, neonatology (NICU), pediatric intensive care (PICU), and pediatric hematology & oncology (in- and outpatient). Gained proficiency in general pediatrics, neonatal resuscitation, critical care management, coordination of oncology follow-up. Developed two clinical decision-support apps deployed in the UZ Gent NICU: bilicurve (bilirubin nomogram) and Fenton growth curves (premature infant growth monitoring).

Sept 2017 – 2021

PhD Fellowship — Research Foundation Flanders (FWO)

Ghent University, Ghent, Belgium

Thesis: "Expanding The Diagnostic Toolbox in Pediatric Oncology: Exploration of the Value of the Cell-Free DNA Methylome."

Promotor: Prof. dr. B. De Wilde · Co-promotor: Prof. dr. K. De Preter

- Patient recruitment, IRB approval, sample processing (plasma prep, DNA isolation, SOP development)
- DNA and RNA sequencing library preparation; qPCR quantification; capillary electrophoresis QC
- Large-scale bioinformatics pipelines (Snakemake / HPC) in Bash, R, Python
- Development of open-source bioinformatics tools

2020 – 2021

Student Supervision

Ghent University

- *"Computational deconvolution of the methylation signal of heterogeneous mixtures: A benchmarking study"* — Master of Bioinformatics, 2021
- *"Deconvolution of Cell-free DNA methylation in Pediatric Cancer"* — Master of Bioinformatics, 2021
- *"Methylation Calling and Tumor Classification from cfDNA of Pediatric and Adult Cancer types"* — Design Project Bioinformatics, 2020
- Jury member: *"Liquid biopsy analysis in neuroblastoma patients"* — Master of Biomedical Sciences, 2020

2017 – 2018

PhD Courses & Training

Various institutions

- Computational Systems Biology in Cancer — Institut Curie, France, 2018
- Advanced Academic English Writing Course — Doctoral Schools, Ghent University, 2018
- Introduction to Deep Learning and Machine Learning — Ghent University & Jülich Supercomputing Center, 2017
- Digital Droplet PCR Course — Doctoral Schools, Ghent University, 2017

June 2013 – 2017

Master in Medicine — *Summa cum laude*

Ghent University

Thesis: "Identification of genetic aberrations predisposing to familial Burkitt lymphoma using whole genome sequencing." Promotor: Prof. Dr. G. Laureys

June 2010 – 2013

Bachelor in Medicine — *Magna cum laude*

Ghent University

PUBLICATIONS

2025

1. **Blood Collection Tube and RNA Purification Method Recommendations for Extracellular RNA Transcriptome Profiling**
   The exRNAQC Consortium · … · **Van Paemel R** · et al.
   Nature Communications, 2025 · DOI: 10.1038/s41467-025-58607-7

2. **A Benchmark of DNA Methylation Deconvolution Methods for Tumoral Fraction Estimation Using DecoNFlow**
   Giuili E · … · **Van Paemel R** · et al.
   bioRxiv (Preprint), 2025 · DOI: 10.1101/2025.11.27.688590

2024

3. **A Fast, Affordable and Minimally-Invasive Diagnostic Test for Cancer of Unknown Primary (CUP) Using DNA Methylation Profiling**
   De Wilde J · **Van Paemel R** · De Koker A · et al.
   Laboratory Investigation, 2024 · DOI: 10.1016/j.labinv.2024.102091

2023

4. **Case Series on Clinical Applications of Liquid Biopsy in Pediatric Solid Tumors**
   Gelineau NU · … · **Van Paemel R** · et al.
   Frontiers in Oncology, 2023 · DOI: 10.3389/fonc.2023.1209150

2022

5. **Cell-free DNA as a Diagnostic and Prognostic Biomarker in Pediatric Rhabdomyosarcoma**
   Lak NSM · van Zogchel LMJ · … · **van Paemel R** · et al.
   JCO Precision Oncology, 2022 · DOI: 10.1200/PO.22.00113

6. **Whole Genome Sequencing and Inheritance-based Variant Filtering as a Tool for Unraveling Missing Heritability in Pediatric Cancer**
   Derpoorter C · **Van Paemel R** · et al.
   Pediatric Hematology and Oncology, 2022 · DOI: 10.1080/08880018.2022.2101723

2021

7. **The Feasibility of Using Liquid Biopsies as a Complementary Assay for Copy Number Aberration Profiling in Routinely Collected Paediatric Cancer Patient Samples**
   Van Paemel R · Vandeputte C · Raman L · et al.
   European Journal of Cancer, 2021 · DOI: 10.1016/j.ejca.2021.09.022

8. **Multisystem Inflammatory Syndrome in Children Related to COVID-19: A Systematic Review**
   Hoste L · **Van Paemel R** · Haerynck F
   European Journal of Pediatrics, 2021 · DOI: 10.1007/s00431-021-03993-5

2020

9. **Genome-wide Study of the Effect of Blood Collection Tubes on the Cell-free DNA Methylome**
   Van Paemel R · De Koker A · Caggiano C · et al.
   Epigenetics, 2020 · DOI: 10.1080/15592294.2020.1827714

10. **When DNA Gets in the Way: A Cautionary Note for DNA Contamination in Extracellular RNA-seq Studies**
    Verwilt J · Trypsteen W · **Van Paemel R** · et al.
    PNAS, 2020 · DOI: 10.1073/pnas.2001675117

11. **Minimally Invasive Classification of Pediatric Solid Tumors Using Reduced Representation Bisulfite Sequencing of Cell-free DNA: A Proof-of-Principle Study**
    Van Paemel R · De Koker A · Vandeputte C · et al.
    Epigenetics, 2020 · DOI: 10.1080/15592294.2020.1790950

2019

12. **The Pitfalls and Promise of Liquid Biopsies for Diagnosing and Treating Solid Tumors in Children: A Review**
    Van Paemel R · Vlug R · De Preter K · et al.
    European Journal of Pediatrics, 2019 · DOI: 10.1007/s00431-019-03545-y

13. **Brain Metastasis and Renal Cell Carcinoma: Prognostic Scores Assessment in the Era of Targeted Therapies**
    El Ali Z · … · **Van Paemel R** · et al.
    Anticancer Research, 2019 · DOI: 10.21873/anticanres.13431

14. **A Versatile Method for Circulating Cell-free DNA Methylome Profiling by Reduced Representation Bisulfite Sequencing**
    Van Paemel R · De Koker A · Vandeputte C · et al.
    bioRxiv (Preprint), 2019

2017

15. **Confirmation of an ARID2 Defect in SWI/SNF-related Intellectual Disability**
    Van Paemel R · De Bruyne P · van der Straaten S · et al.
    American Journal of Medical Genetics Part A, 2017 · DOI: 10.1002/ajmg.a.38407

PRESENTATIONS

2026	Lymphadenopathy work-up Master of Specialist Medicine – Pediatrics
2025	Unmasking Familial Hemophagocytic Lymphohistiocytosis type 3: Chronic Disseminated Rubella Infection Following MMR Vaccination *(Travel grant)* eBMT Inborn Error Working Party
2025	Acute management of sickle cell anemia National Meeting - Pediatrics

2019	Non-invasive classification of pediatric tumor types using reduced representation bisulfite sequencing of cell-free DNA: a proof-of-principle study EACR-ESMO Joint Conference on Liquid Biopsies · Bergamo, Italy

ACADEMIC POSTERS

2026	Mycobacterium abscessus as a Complication of Immunosuppression after Allogeneic Stem Cell Transplantation for Congenital Neutropenia eBMT · Madrid, Spain
2025	Ten Years of CAR-T Cell Therapy: Real-world Experience from the Belgian Cohort INSPIRED · Chicago, USA
2025	Fatal HLH as a Complication of Secondary Histiocytic Sarcoma Treated with Trametinib and Chemotherapy CCLS · Budapest, Hungary
2019	Minimally-Invasive Classification of Pediatric Solid Tumors Using Reduced Representation Bisulfite Sequencing of Cell-free DNA: A Proof-of-Principle Study SIOP 2019 · Lyon, France

SOFTWARE & TOOLS

clinAnnotR	Harmonise treatment and lab values of multiple cases on one figure · R github.com/rmvpaeme/clinAnnotR
EU Pediatric Trial Monitor	Interactive EU clinical trial dashboard · R/Shiny github.com/rmvpaeme/shiny_trials
fenton-shiny	Fenton growth curve visualiser — deployed at UZ Gent NICU · R/Shiny github.com/rmvpaeme/fenton-shiny
bilicurve-shiny	Bilirubin nomogram app — deployed at UZ Gent NICU · R/Shiny
Stolling	Interactive overview of the clotting cascade · HTML/JS rmvpaeme.github.io/stolling

AWARDS & CERTIFICATES

Awards

Award for Oral Presentation — Oncopoint, 2019

Award for Flash Presentation — Institut Curie, Paris, 2018

Best Poster by Young Investigator — BeSHG, 2018

Award for Storm Session Presentation — Oncopoint, 2018

Certificates

Electrocardiography

Newborn Life Support (NLS) Provider

European Pediatric Advanced Life Support (EPALS) Provider

European Board of Paediatricians (EPB)-Certified

GCP ICH E6(R3) Certified